

# CORE (1) - Mathematics PT-2 Study Material

Concept-complete teaching, learner-conditioned depth

|          |       |           |
|----------|-------|-----------|
| CORE (1) | STUDY | GROUNDING |
|----------|-------|-----------|

## Purpose

This is the teaching book. It does not merely summarize formulas. It rebuilds the mathematics from meaning -> representation -> formal rule -> worked example -> guided practice -> independent check. The PT-2 questions determine the required scope; the learner's 9/30 working determines the depth, order and bridges.

Grounding: original PT-2 Mathematics paper (28-08-2026) + marked 9/30 learner working. Internal source codes: [PT2-Qx] and [SW-Qx].

## How to read the badges and citations

PT-2 Qx

RECONSTRUCT

VERIFY

LEARNER TRAP

### Source codes

[PT2-Qx] means the original printed PT-2 item. [SW-Qx] means the learner's marked working on that item. A lesson may cite both because the question tells us what must be taught, while the working tells us where the explanation must go deeper.

## Question -> study map

| PT-2 | Core (1) lesson | What is being learned               | Why extra attention?                                   |
|------|-----------------|-------------------------------------|--------------------------------------------------------|
| Q1   | B1              | Axiom vs postulate                  | Mostly consolidation.                                  |
| Q2   | C2              | Distance formula                    | Formula recognized; rebuild why it works for transfer. |
| Q3   | A2/C1           | Signs and quadrants                 | Wrong quadrant: teach sign transformation.             |
| Q4   | D1              | y-intercept                         | Preserve existing strength.                            |
| Q5   | D2              | Line from slope + point             | Needs full reconstruction of $y = mx + c$ .            |
| Q6   | F1              | Counting pairs                      | Teach why $n(n-1)/2$ counts pairs.                     |
| Q7   | C3              | Collinearity and slope              | Definition + data-copy error + method.                 |
| Q8   | B2              | Euclid V / parallel lines           | Fragments remembered but meaning contradicted.         |
| Q9   | G1              | Information sufficiency             | Printed item is underdetermined.                       |
| Q10  | A1/D3           | Systems + line through intersection | Multi-stage reasoning and verification.                |
| Q11  | C2/E1           | Equidistant point                   | Needs complete teaching.                               |
| Q12  | A3/E2           | Equilateral coordinate geometry     | Good geometry start; algebra expansion breaks.         |
| Q13  | C3/D4/F2        | Slope from data + linear model      | Delta-y / delta-x roles unstable.                      |
| Q14  | A1/D3/F3        | River-current model + systems       | Model good; expression meaning lost.                   |

Study order: A (algebra engine) -> B/C/D (geometry and lines) -> E/F/G (construction, modelling and judgement). Do not skip A even though it is not a named chapter on the paper; it is the upstream prerequisite repeatedly breaking later solutions.

# A1 Equality is a promise: legal transformations

PT-2 Q10 • Q14

RECONSTRUCT

A1

**You should leave this lesson able to: explain why an equation remains true only when the same valid operation is applied to both sides, and preserve compound expressions such as  $x + y$ .**

Grounding / linkage: [PT2-Q10,Q14] | [SW-Q10,Q14] | Concept: A1

## Big idea

The equals sign does not mean “now calculate the next line.” It means the expression on the left and the expression on the right have the same value. Every legal algebra step must preserve that promise.

Think of  $x + y = 24$  as one statement about a combined quantity. If  $x$  is the boat speed and  $y$  is the current speed,  $x + y$  is the downstream speed. Removing  $y$  from the left without doing anything to the right changes the statement; it is not simplification, it is a new and unsupported claim.

## How to move from one line to the next

1. Name the whole expression you are changing. Example:  $x + y$  is the downstream-speed expression.
2. Choose a legal operation. Example: subtract  $y$  from both sides.
3. Write the operation on both sides:  $x + y - y = 24 - y$ .
4. Simplify only after the legal operation:  $x = 24 - y$ .
5. Check by substituting a possible pair, such as  $x=20$ ,  $y=4$ .

## Why the learner's Q14 route failed

The original work correctly formed combined speeds, but then treated the value of  $x+y$  as if it were the value of  $x$  alone. Core (1) therefore keeps the modelling strength and repairs the algebraic meaning-preservation step.

## Worked example

Solve  $2a + b = 17$  for  $a$  in terms of  $b$ .

$$2a + b = 17$$

$$2a + b - b = 17 - b \quad (\text{subtract } b \text{ from both sides})$$

$$2a = 17 - b$$

$$a = (17 - b)/2 \quad (\text{divide both sides by } 2)$$

## Guided practice

1. From  $p + q = 31$ , express  $p$  in terms of  $q$ .
2. From  $3x - 5 = 19$ , solve for  $x$ .
3. Is  $x + y = 24 \rightarrow x = 24$  legal? Explain in one sentence.

## Check after you try

1)  $p = 31 - q$ ; 2)  $x = 8$ ; 3) No;  $y$  was removed only from the left side.

## Say it aloud

“I am not moving symbols across the equals sign. I am applying the same operation to both sides.”

# A2 Signs, substitution and transformed coordinates

PT-2 Q3 • Q10

APPLY

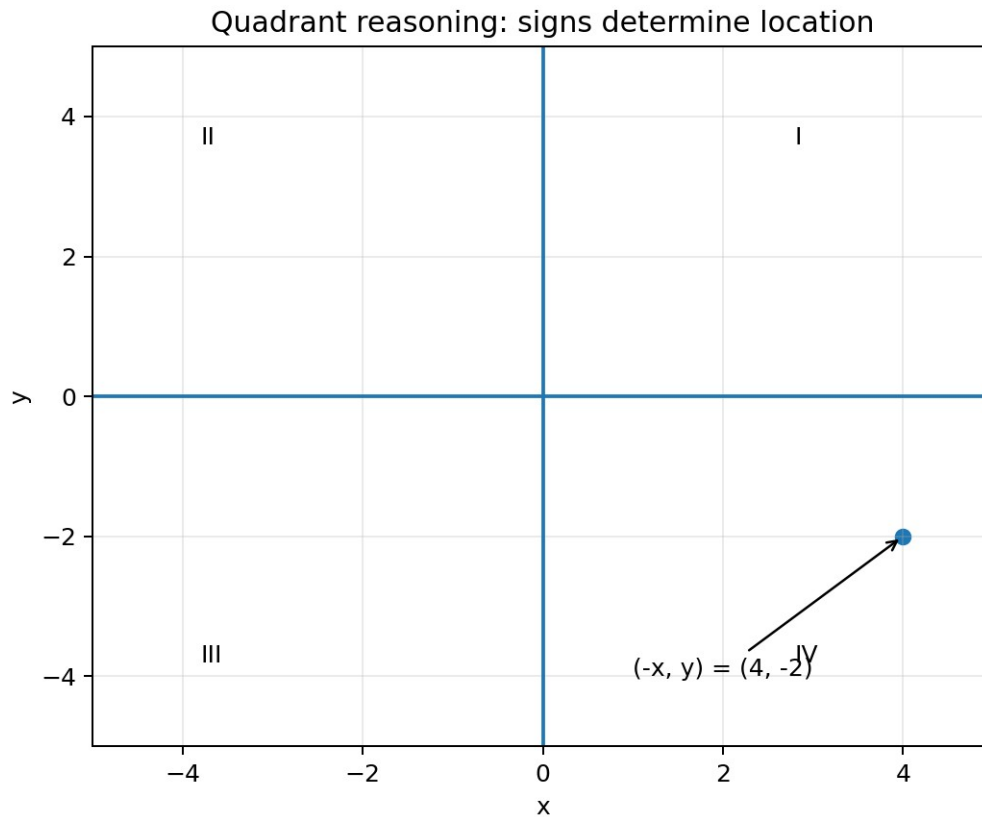
A2

**You should leave this lesson able to: track the sign of a number after substitution, especially when a variable is already negative.**

Grounding / linkage: [PT2-Q3,Q10] | [SW-Q3,Q10] | Concept: A2

### Big idea

A minus sign in front of a negative number changes its sign. If  $x < 0$ , then  $-x > 0$ . Do not decide a quadrant from the letters alone; first determine the signs of the actual coordinates.



Visual guide: use a sample negative value to reason about  $(-x, y)$ .

### Build the reasoning

Suppose  $x = -4$  and  $y = -2$ . Both satisfy  $x < 0$  and  $y < 0$ . Then  $(-x, y) = (-(-4), -2) = (4, -2)$ . The first coordinate is positive and the second is negative, so the point lies in Quadrant IV.

### Do not memorize the answer to Q3

The transferable method is: determine the sign of each coordinate after every transformation, then use the quadrant sign pattern.

### Guided practice

1. If  $a < 0$  and  $b > 0$ , what are the signs of  $(-a, b)$ ?
2. If  $m > 0$  and  $n < 0$ , what are the signs of  $(-m, -n)$ ?
3. Give one numerical example for  $(-x, y)$  when  $x < 0, y < 0$ .

### Check after you try

1)  $(+, +)$ , Quadrant I; 2)  $(-, +)$ , Quadrant II; 3) Example:  $x = -5, y = -3 \rightarrow (5, -3)$ .

## A3 Expand squares without corrupting the expression

PT-2 Q12

RECONSTRUCT

A3

**You should leave this lesson able to: derive and use  $(a-b)^2$  correctly, then recognize where each term comes from.**

Grounding / linkage: [PT2-Q12] | [SW-Q12] | Concept: A3

### Big idea

Do not treat  $(x-3)^2$  as “square each visible number somehow.” It means multiply the entire bracket by itself:  $(x-3)(x-3)$ .

$$(x-3)^2 = (x-3)(x-3)$$

$$= x^2 - 3x - 3x + 9$$

$$= x^2 - 6x + 9$$

### Learner trap from Q12

The original working effectively produced an 81-type constant. The correct constant is  $3^2 = 9$ . There is no second squaring of 9.

### Where the identity comes from

For any  $a$  and  $b$ :  $(a-b)^2 = (a-b)(a-b) = a^2 - ab - ab + b^2 = a^2 - 2ab + b^2$ .

### Faded practice

1.  $(y-5)^2$
2.  $(2x-3)^2$
3.  $(p+4)^2$
4. Expand  $(x-3)^2 + (y-\sqrt{3})^2$  without simplifying beyond expanded terms.

### Check after you try

1)  $y^2-10y+25$ ; 2)  $4x^2-12x+9$ ; 3)  $p^2+8p+16$ ; 4)  $x^2-6x+9 + y^2-2\sqrt{3}y+3$

### Connection forward

You will use this algebra inside coordinate geometry in Lesson E2. The geometry can be correct while the algebra is wrong; we keep those capabilities separate.

## B1 Euclid: axioms, postulates and what they mean

PT-2 Q1

DISCRIMINATE

B1

**You should leave this lesson able to: distinguish a general axiom from a geometry-specific postulate and classify the statements in Q1.**

Grounding / linkage: [PT2-Q1] | [SW-Q1] | Concept: B1

### Concept first

Axioms are general accepted truths used across mathematics. Postulates are assumptions stated for geometry. Q1 is testing classification, not calculation.

**Axiom:** Things equal to the same thing are equal to one another.

**Axiom:** If equals are added to equals, the wholes are equal.

**Axiom:** If equals are subtracted from equals, the remainders are equal.

**Postulate:** A circle can be drawn with any centre and any radius.

### Why option (c) is the odd one out

The circle statement is a geometry construction postulate, not a common axiom. That is why it is “not an axiom.”

## B2 Euclid’s Fifth Postulate: why parallel lines are about angle sums

PT-2 Q8

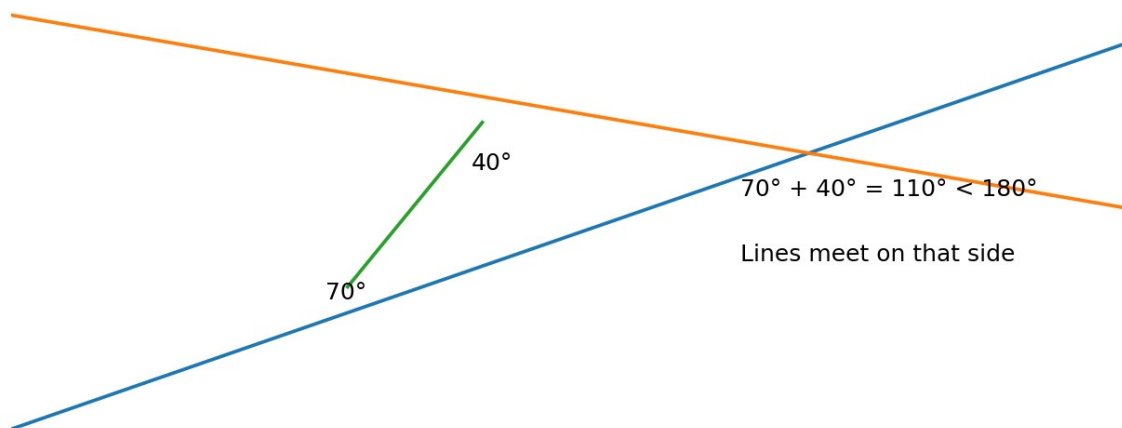
EXPLAIN

B2

**You should leave this lesson able to: explain in words and with a diagram why same-side interior angles determine whether two lines meet.**

Grounding / linkage: [PT2-Q8] | [SW-Q8] | Concept: B2

Euclid V: same-side interior angles control meeting/parallelism



Visual guide: a sum below 180 degrees means the lines meet on that side.

### The meaning

When a transversal cuts two lines, look at the interior angles on the same side. If their sum is less than two right angles (180 degrees), Euclid V says the two lines meet on that side when extended.

Therefore, if the same-side interior angles add to exactly 180 degrees, the lines do not meet on either side: this gives the familiar parallel-line condition in Euclidean geometry.

### Example

70 degrees + 40 degrees = 110 degrees, which is less than 180 degrees. So those lines meet; they are not parallel.

#### Learner trap from Q8

The original answer correctly recognized “sum less than 180 degrees -> meet” in one place, but then wrote that this represented parallel lines. The repair is to tie the words “meet” and “parallel” to one clear picture.

### Explain, do not just state

1. What happens if the same-side interior angles total 150 degrees?
2. What angle-sum condition corresponds to parallel lines?
3. Write a two-sentence explanation suitable for Q8.

#### Check after you try

1) The lines meet on that side.; 2) 180 degrees.; 3) Use the model explanation above in your own words.

## C1 Coordinate plane: position comes from two signed coordinates

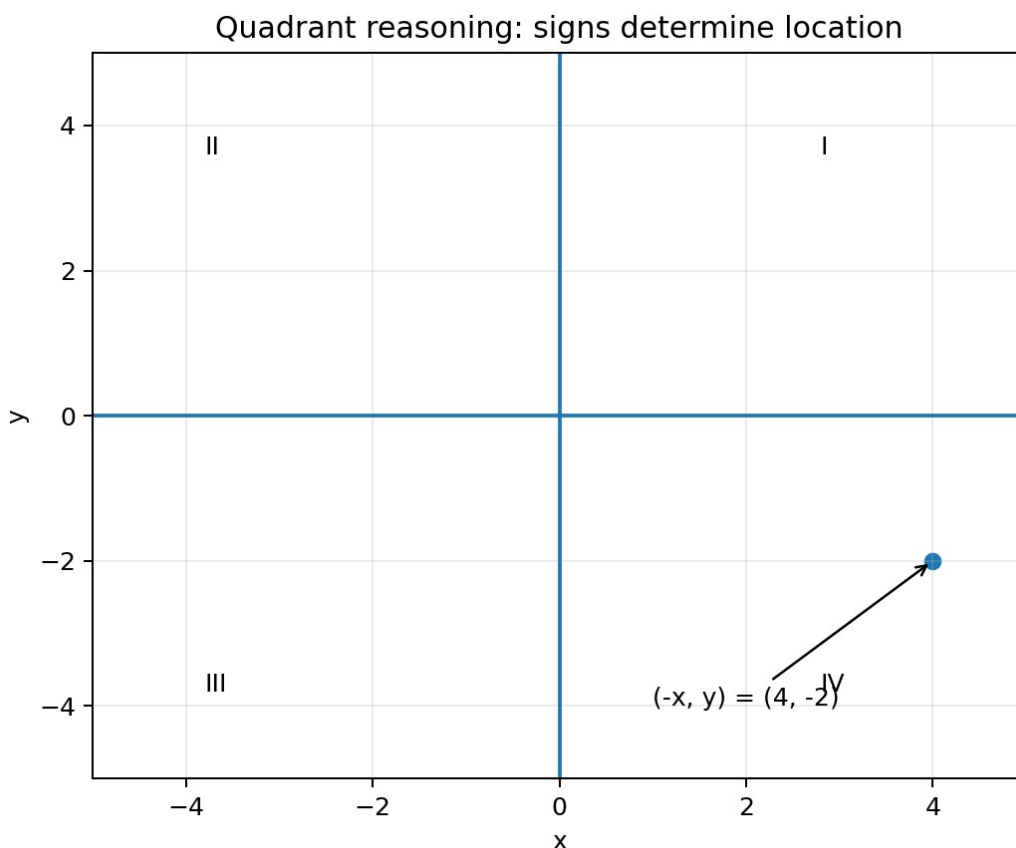
PT-2 Q3

REPRESENT

C1

**You should leave this lesson able to: read a point as horizontal position first, vertical position second, and use signs rather than guesswork.**

Grounding / linkage: [PT2-Q3] | [SW-Q3] | Concept: C1



The coordinate plane is a map of two signed directions.

An ordered pair  $(x, y)$  tells you: move  $x$  units horizontally from the origin, then  $y$  units vertically.  $x > 0$  means right;  $x < 0$  means left.  $y > 0$  means up;  $y < 0$  means down.

### Memory anchor

Quadrants: I (+,+), II (-,+), III (-,-), IV (+,-). But always work out the signs first.

## C2 Distance formula is Pythagoras in coordinates

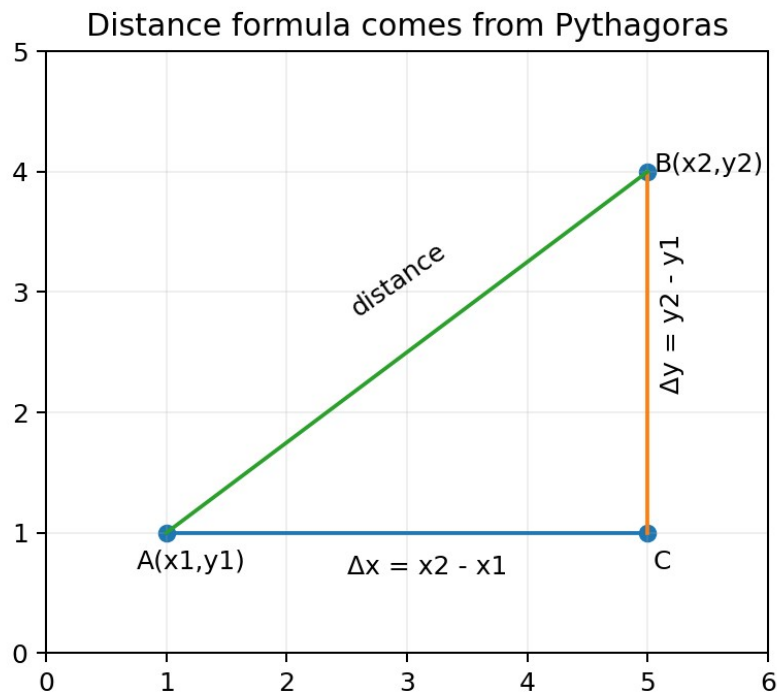
PT-2 Q2 • Q11 • Q12

RECONSTRUCT

C2

**You should leave this lesson able to: derive the distance formula from horizontal and vertical changes and use squared distance when equality is enough.**

Grounding / linkage: [PT2-Q2,Q11,Q12] | [SW-Q2,Q11,Q12] | Concept: C2



Visual guide: the horizontal and vertical coordinate changes form a right triangle.

For  $A(x_1, y_1)$  and  $B(x_2, y_2)$ , the horizontal leg has length  $|x_2 - x_1|$  and the vertical leg has length  $|y_2 - y_1|$ .  
Pythagoras gives:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

### Why squared distance is sometimes better

If a problem says two distances are equal,  $PA = PB$ , you may square both sides and work with  $PA^2 = PB^2$ . This removes square roots and makes the algebra cleaner without changing the equality because distances are non-negative.

#### Guided practice

1. Find the distance between (1, 2) and (5, 5).
2. Write  $PA^2$  if  $P=(a, 0)$  and  $A=(7, 6)$ .
3. Why can Q11 use squared distances?

#### Check after you try

1) 5; 2)  $(a-7)^2 + 36$ ; 3) Because  $PA=PB$  is equivalent to  $PA^2=PB^2$  for non-negative distances.

## C3 Slope and collinearity: keep x with x and y with y

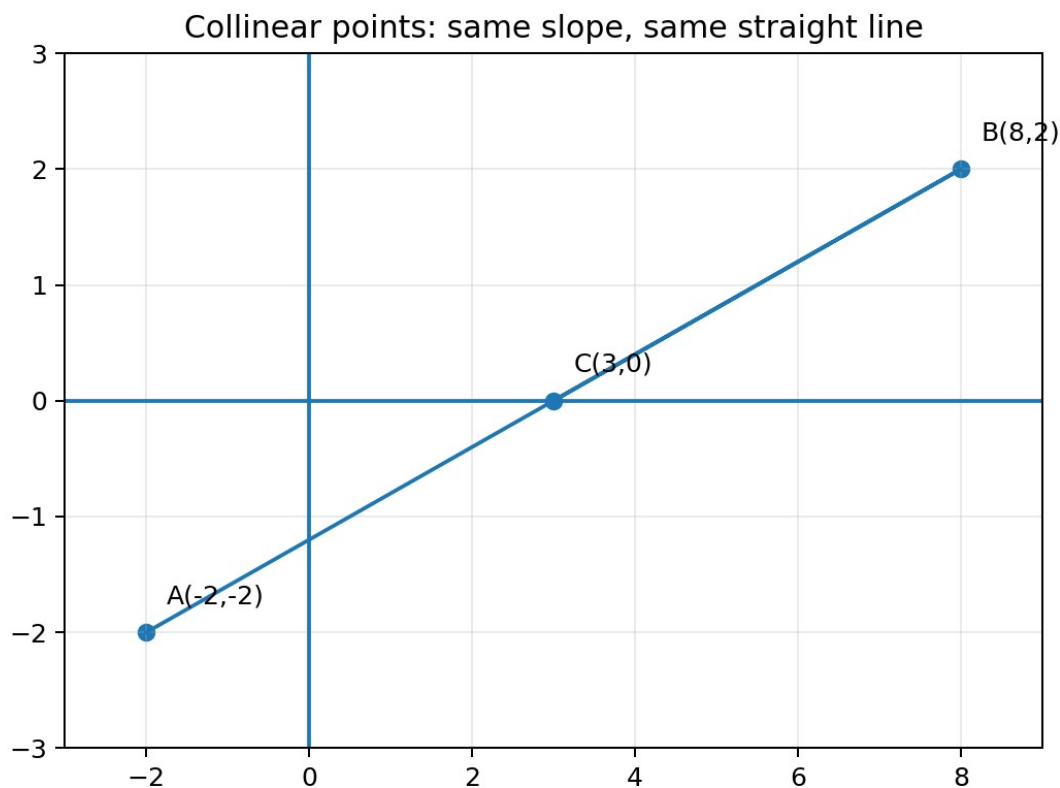
PT-2 Q7 • Q13

CONNECT

C3

**You should leave this lesson able to: compute slope as vertical change divided by horizontal change, interpret its units, and use equal slope to test collinearity.**

Grounding / linkage: [PT2-Q7, Q13] | [SW-Q7, Q13] | Concept: C3



The three PT-2 points lie on one straight line because their slopes agree.

### Slope meaning

$m = (\text{change in } y)/(\text{change in } x) = (y_2 - y_1)/(x_2 - x_1)$ . "Vertical over horizontal" is a meaning, not just a formula.

For A(-2, -2), B(8, 2):  $m_{AB} = (2 - (-2))/(8 - (-2)) = 4/10 = 2/5$ .

For A(-2, -2), C(3, 0):  $m_{AC} = (0 - (-2))/(3 - (-2)) = 2/5$ . Equal slopes mean the direction from A to B and from A to C is the same, so the points are collinear.

### Data-copy routine from the original work

The learner copied B(8, 2) as approximately B(3, 2). Before calculating: **Read -> Copy -> Point back -> Check**. One copied coordinate can create a completely different mathematical problem.

### Slope units

In Q13,  $x$  is selling price (rupees per litre) and  $y$  is weekly demand (litres). Therefore slope units are litres per rupee. Units help you detect whether you accidentally used  $\Delta x / \Delta y$ .

### Guided practice

1. Find slope through (2, 3) and (6, 11).
2. If  $x$  is time (s) and  $y$  is distance (m), what are slope units?
3. What test can establish that three non-vertical points are collinear?

### Check after you try

- 1) 2; 2) m/s; 3) Compare the slopes of two segments sharing a point.

## D1 Intercepts: where a line meets an axis

PT-2 Q4

APPLY

D1

You should leave this lesson able to: find a y-intercept by using the defining condition  $x=0$ .

Grounding / linkage: [PT2-Q4] | [SW-Q4] | Concept: D1

### Meaning before rule

Every point on the y-axis has  $x=0$ . So the y-intercept of an equation is found by setting  $x=0$  because you are asking where the line reaches that axis.

PT-2 mechanism:  $5x + 3y = 4$ . Set  $x=0$ :  $3y=4$ , so  $y=4/3$ . The intercept point is  $(0, 4/3)$ .

### Quick transfer

1. Find the y-intercept of  $2x + 5y = 20$ .
2. Find the x-intercept of the same line.

### Check after you try

1)  $(0, 4)$ ; 2)  $(10, 0)$

## D2 A straight line from its slope and one point

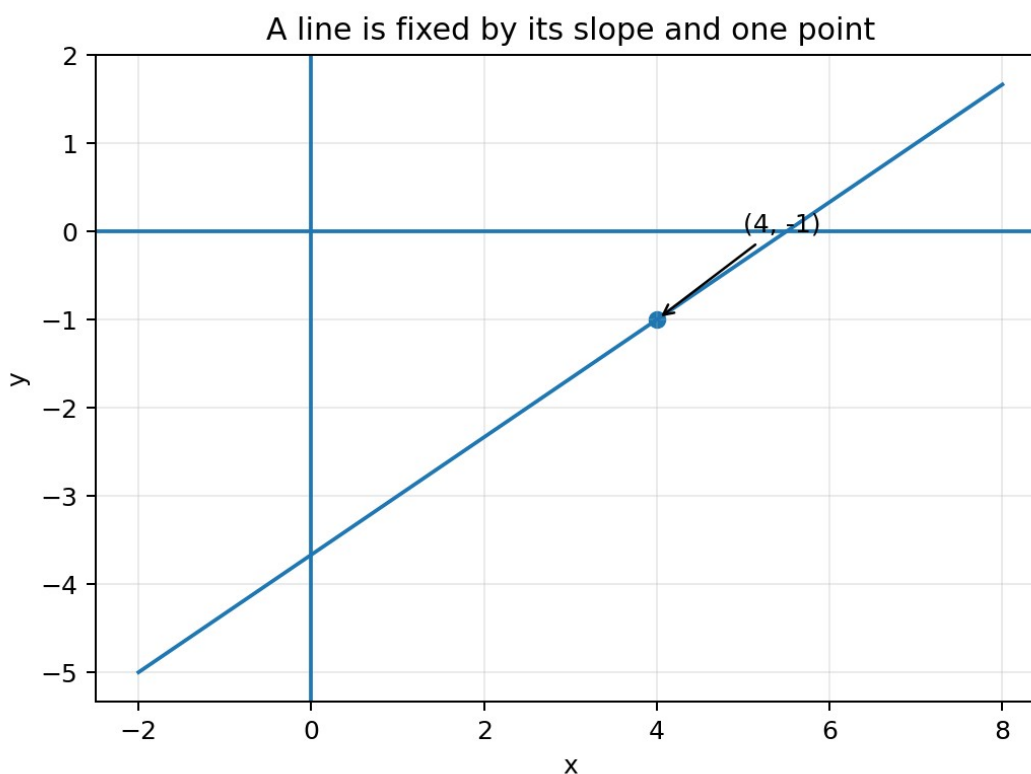
PT-2 Q5

RECONSTRUCT

D2

You should leave this lesson able to: understand  $y=mx+c$ , use a known point to solve for  $c$ , and verify that the point lies on the resulting line.

Grounding / linkage: [PT2-Q5] | [SW-Q5] | Concept: D2



Visual guide: slope fixes the direction; one point fixes which parallel line it is.

In  $y = mx + c$ ,  $m$  controls how much  $y$  changes when  $x$  increases by 1.  $c$  is the  $y$ -intercept. If  $m$  is already known, one point is enough to determine  $c$ .

### Build the line

1. Write  $y = mx + c$ .
2. Insert the known slope  $m$ .
3. Substitute the coordinates of the known point.
4. Solve the resulting one-variable equation for  $c$ .
5. Verify the point in your final line equation.

### Worked example

$m = 2/3$  and point  $(4, -1)$ :  $-1 = (2/3)(4) + c = 8/3 + c$ , so  $c = -11/3$ . Hence  $y = (2/3)x - 11/3$ .

### Verification

Put  $x=4$  into the final equation:  $y = 8/3 - 11/3 = -1$ . The given point is recovered.

### Faded practice

1. Find the line with slope 3 passing through  $(2, 5)$ .
2. Find the line with slope  $-1/2$  through  $(4, 1)$ .

### Check after you try

1)  $y=3x-1$ ; 2)  $y=-(1/2)x+3$

## D3 Simultaneous equations: solve, verify, then reuse the point

PT-2 Q10 • Q14

CONNECT

D3

**You should leave this lesson able to: solve a pair of linear equations reliably and keep the solution as a state that can feed the next stage of a problem.**

Grounding / linkage: [PT2-Q10,Q14] | [SW-Q10,Q14] | Concept: D3

### Why this matters

Q10 is not one calculation. It is a chain: solve the intersection  $\rightarrow$  store the point  $\rightarrow$  combine it with  $(2, -3)$   $\rightarrow$  find slope  $\rightarrow$  construct the required line. If an early state is wrong, every downstream step is contaminated.

### Multi-stage state discipline

1. Normalize the equations into a clear form.
2. Eliminate one variable by adding or subtracting.
3. Solve for the remaining variable.
4. Substitute back to get the ordered pair.
5. Verify the pair in both original equations.
6. Only then reuse that point in the second stage.

### Worked example using the PT-2 mechanism

$$x + y + 4 = 0 \rightarrow x + y = -4$$

$$3x - y - 8 = 0 \rightarrow 3x - y = 8$$

Add:  $4x = 4 \rightarrow x = 1$ . Then  $y = -5$ . Verify both equations.

Now the second stage begins. Points (1,-5) and (2,-3) give slope  $m = 2$ . Point-slope form gives  $y+5 = 2(x-1)$ , so  $y = 2x - 7$ .

### Mandatory habit

Every simultaneous-equation answer ends with a two-equation substitution check. Treat this as part of the solution, not optional decoration.

### Guided practice

1. Solve  $x+y=9$  and  $x-y=3$ .
2. Verify the result in both equations.
3. Find the line through that solution point and (4,1).

### Check after you try

1) (6,3); 2)  $6+3=9$  and  $6-3=3$ ; 3) Slope  $= (1-3)/(4-6)=1$ ; line  $y=x-3$

## D4 Slope from real data: roles, units and prediction

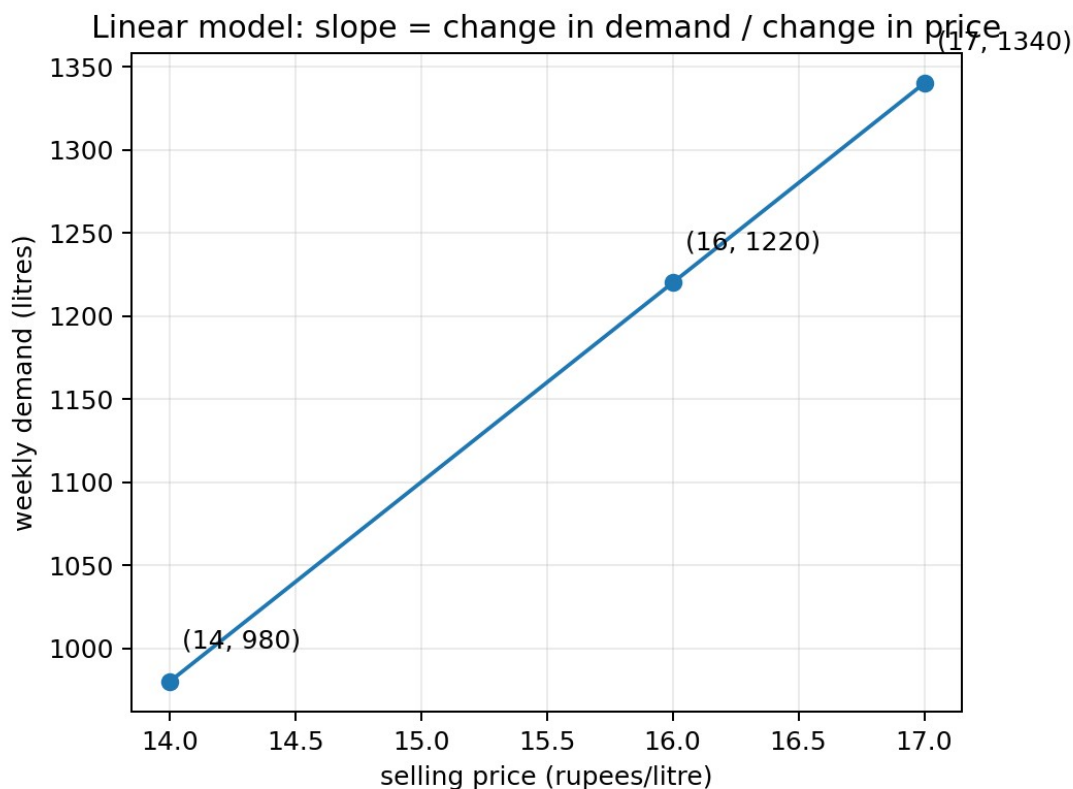
PT-2 Q13

MODEL

D4

**You should leave this lesson able to: assign  $x$  and  $y$  deliberately, compute slope with units, and extend a linear model one step.**

Grounding / linkage: [PT2-Q13] | [SW-Q13] | Concept: D4



The line through two observed data points defines the stated linear model.

Let  $x$  = selling price and  $y$  = weekly demand. The data points are (14,980) and (16,1220).

$m = (1220-980)/(16-14) = 240/2 = 120$  litres per rupee.

From price 16 to price 17 is an increase of 1 rupee, so the linear model adds 120 litres:  $1220 + 120 = 1340$  litres.

**Reality check vs mathematical task**

In real markets, demand often decreases when price rises. But the printed question explicitly asks us to assume a linear relationship using the two supplied points. Solve the stated mathematical model while recognizing that a model can be unrealistic.

## E1 Equidistant point on the x-axis

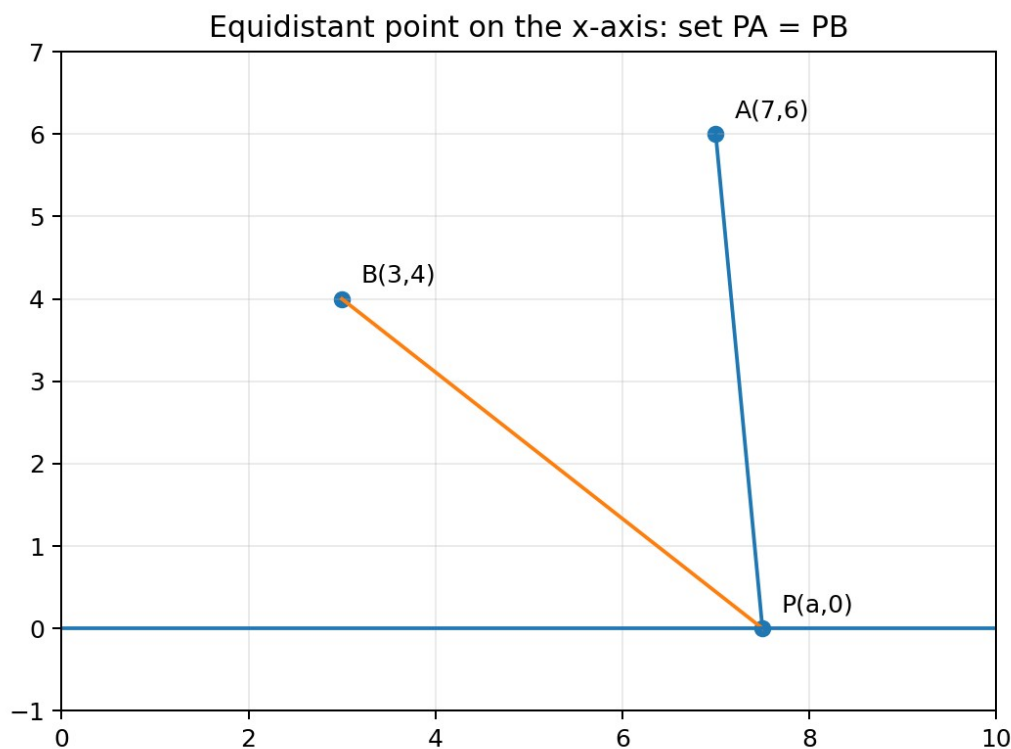
PT-2 Q11

MODEL

E1

You should leave this lesson able to: translate “on the x-axis” and “equidistant” into coordinates and equations.

Grounding / linkage: [PT2-Q11] | [SW-Q11] | Concept: E1



The unknown point is constrained to  $y=0$ , so write  $P(a,0)$ .

“On the x-axis” gives a representation immediately:  $P(a,0)$ . “Equidistant from A and B” means  $PA = PB$ . This is modelling before algebra.

Use squared distances:  $(a-7)^2 + (0-6)^2 = (a-3)^2 + (0-4)^2$ .

Expand:  $(a-7)^2 + 36 = (a-3)^2 + 16$ . After cancellation and simplification:  $a = 15/2$ . Therefore  $P = (15/2, 0)$ .

### Why this lesson connects backward

This problem needs C2 (distance) and A3 (expanding squares). If the model is correct but the expansion fails, repair A3 instead of reteaching coordinate geometry from zero.

## E2 Equilateral triangle in coordinates: one side, two possible third vertices

PT-2 Q12

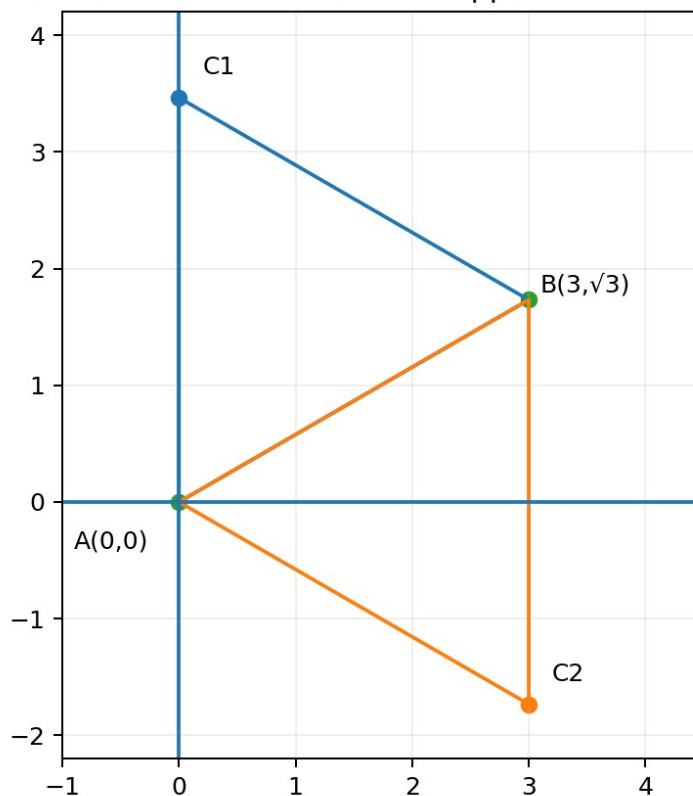
TRANSFER

E2

You should leave this lesson able to: represent the equal-side condition with distances, solve the system and understand why two geometric solutions exist.

Grounding / linkage: [PT2-Q12] | [SW-Q12] | Concept: E2

Two valid third vertices lie on opposite sides of AB



Two equilateral triangles can be built on opposite sides of the same segment AB.

$A=(0,0)$ ,  $B=(3,\sqrt{3})$ . First compute  $AB^2 = 3^2 + (\sqrt{3})^2 = 12$ . This tells us the target squared side length without introducing square roots later.

Let  $C=(x,y)$ . Equilateral means  $AC^2 = 12$  and  $BC^2 = 12$ :

$$x^2 + y^2 = 12$$

$$(x-3)^2 + (y-\sqrt{3})^2 = 12$$

### Solve carefully

Expand the second equation:  $x^2 + y^2 - 6x - 2\sqrt{3}y + 12 = 12$ . Using  $x^2 + y^2 = 12$  gives  $3x + \sqrt{3}y = 6$ .

So  $y = \sqrt{3}(2-x)$ . Substitute in  $x^2 + y^2 = 12$ :  $x^2 + 3(2-x)^2 = 12 \rightarrow 4x(x-3) = 0$ . Thus  $x=0$  or  $x=3$ .

If  $x=0$ ,  $y=2\sqrt{3}$ . If  $x=3$ ,  $y=-\sqrt{3}$ . Therefore both  $C=(0,2\sqrt{3})$  and  $C=(3,-\sqrt{3})$  are valid.

### Assessment-quality note

The printed question says “find the third vertex” but two third vertices are geometrically valid unless an orientation is specified. Core (2) accepts both.

### Explain the geometry

1. Why are two answers possible?
2. Which earlier algebra lesson is critical here?
3. What should you verify after finding a candidate vertex?

### Check after you try

- 1) The third point may lie on either side of AB.;
- 2) A3 expansion of squares.;
- 3) All three squared side lengths are equal.

## F1 Counting pairs: why six points give fifteen lines

PT-2 Q6

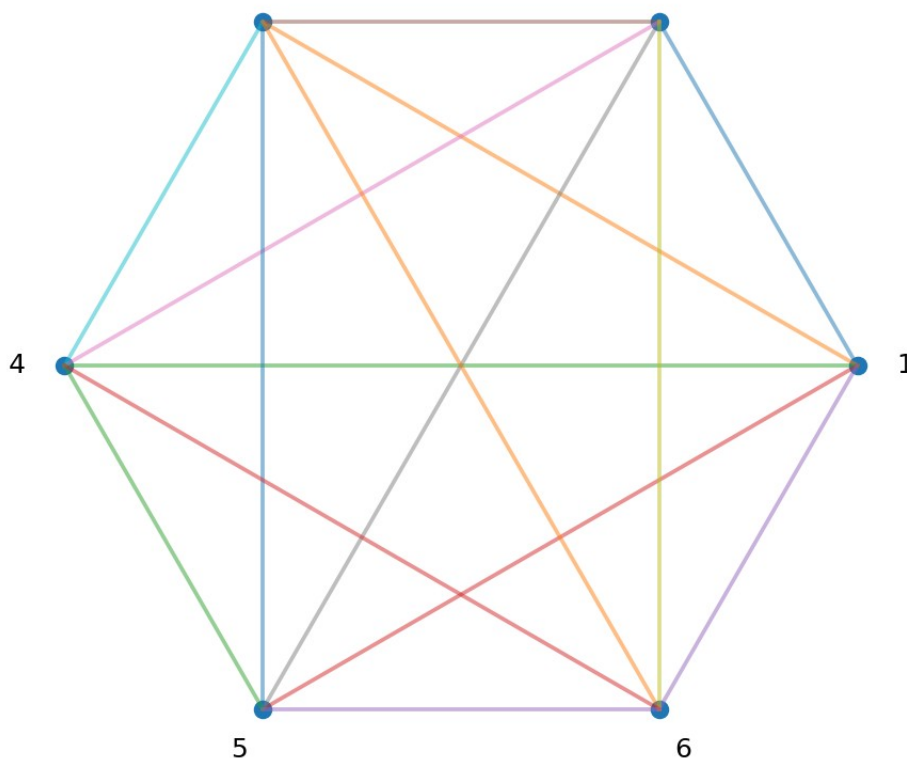
RECONSTRUCT

F1

**You should leave this lesson able to: see that each line is determined by a pair of points and count each unordered pair exactly once.**

Grounding / linkage: [PT2-Q6] | [SW-Q6] | Concept: F1

Six points: each pair determines one line (15 pairs)



Every connection corresponds to one pair of points.

Choose one point: it connects to 5 others. If you repeat this for all 6 points, you count every line twice - once from each endpoint. So  $6 \times 5 / 2 = 15$ .

This is the same as the combination count  $C(6,2)$ . The condition “no three are collinear” ensures no two different pairs collapse into the same line through a third point.

### Quick transfer

1. How many lines from 5 points if no three are collinear?
2. Why does “no three collinear” matter?

### Check after you try

1) 10; 2) Because otherwise multiple pairs could describe the same line.

## F2 Linear models from two data points

PT-2 Q13

TRANSFER

F2

**You should leave this lesson able to: build and interpret a linear model from two observations, not just compute a fraction.**

Grounding / linkage: [PT2-Q13] | [SW-Q13] | Concept: F2

### Model statement

A straight-line model assumes the same change in  $y$  for each equal change in  $x$ . Once two points are given, the slope fixes the rate and one point fixes the line.

For Q13, the slope is +120 litres per rupee. Starting from (16,1220), moving to  $x=17$  means one step of +1 in  $x$ , so apply one slope step: +120 in  $y$ .

### Connection to D4

D4 teaches the mechanics and units of slope. F2 emphasizes the model: what assumption is being made and how to extend it.

## F3 River-current modelling: translate words into relative speeds

PT-2 Q14

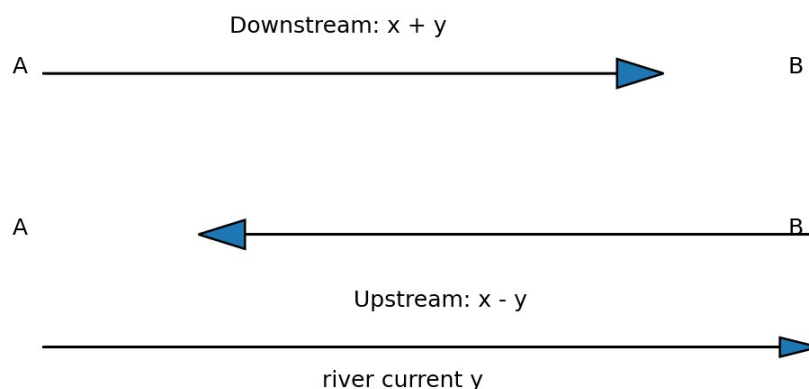
MODEL

F3

**You should leave this lesson able to: build downstream and upstream speeds from still-water speed and current speed, then solve the resulting system.**

Grounding / linkage: [PT2-Q14] | [SW-Q14] | Concept: F3

Boat + current: direction changes the sign of  $y$



With the current, speeds add; against the current, the current subtracts from the boat speed.

Let  $x$  = speed of the boat in still water and  $y$  = speed of the river current.

Downstream: boat and current act in the same direction, so effective speed =  $x+y$ . Upstream: they oppose, so effective speed =  $x-y$ .

The distance is 48 km. Downstream time 2 h gives  $x+y = 48/2 = 24$ . Upstream time 3 h gives  $x-y = 48/3 = 16$ .

Add the equations:  $2x=40 \rightarrow x=20$ . Substitute:  $y=4$ .

### Learner strength to preserve

The original answer sheet already showed the correct idea  $x+y$  and  $x-y$ . The repair is not “learn river problems from zero”; it is “keep the model intact while solving the equations.”

# G1 Mathematical judgement: is there enough information?

PT-2 Q9

DIAGNOSE

G1

**You should leave this lesson able to: recognize when a printed problem does not determine one unique numerical answer.**

Grounding / linkage: [PT2-Q9] | [SW-Q9] | Concept: G1

## Important

A mathematics question is not automatically well-posed just because it is printed on a test. Good mathematical reasoning includes checking whether the data actually determine the requested quantity.

Q9 gives  $2x + cy = 8$  and says the solution has equal  $x$  and  $y$ . Put  $y=x$ :  $2x + cx = 8$ , so  $x(2+c)=8$ .

Therefore  $c = 8/x - 2$ . Because  $x$  itself is not specified, many values of  $c$  are possible, paired with different equal-coordinate solutions. There is no single numerical  $c$  from the printed information.

## How to write this responsibly

"From  $x=y$ ,  $c=8/x-2$ . Since  $x$  is not given,  $c$  is not uniquely determined from the printed data." This is a mathematically legitimate conclusion.

# H1 Before Core (2): readiness checklist

PT-2 Q1-Q14

VERIFY

H1

**You should leave this lesson able to: decide whether you can now reconstruct the tested methods without looking at worked solutions.**

Grounding / linkage: [PT2-Q1-Q14] | Concept: H1

- ☐ I can explain why the distance formula is Pythagoras, not just recite it.
- ☐ I can preserve equality from line to line and verify simultaneous-equation solutions.
- ☐ I can expand  $(a-b)^2$  from multiplication.
- ☐ I can explain collinear vs coplanar and use slope to test collinearity.
- ☐ I can construct  $y=mx+c$  from a slope and point.
- ☐ I can solve a two-stage line-through-intersection problem.
- ☐ I can model downstream/upstream speeds without confusing  $x+y$  with  $x$ .
- ☐ I can recognize when a question is underdetermined.

## Next step

Use Core (2) as an attempt-first transfer booklet. The actual PT-2 question image appears first. The support page immediately after each question contains H1, H2, H3, a full reasoning route, the solution and an exact Core (1) repair link.

## Source key

[PT2-Qx] = original Periodic Test-2 Mathematics question x shown in the attached paper. [SW-Qx] = the learner's marked answer-sheet evidence for that question. These citations are internal evidence lineage, not external publication references.